Assessment Inequity and Land Value Taxation: A Multi-City Analysis of Property Tax Disparities in Cook County, New York City, and Philadelphia

Krishna Singh, Isabelle Byrne
Northeastern University
Boston, United States

***Abstract*—Property tax assessments in the United States are widely documented to be regressive: lower-value properties bear higher effective tax rates relative to market value, with the burden falling disproportionately on lower-income and majority-Black neighborhoods. We ask two questions. Do official assessed values in Cook County (IL), New York City (NY), and Philadelphia (PA) exhibit systematic disparities relative to model-predicted market values in these neighborhoods? And would a revenue-neutral land-value-only tax (LVT) – total collections held constant by adjusting the rate applied to the land-only base – reduce those disparities? We train XGBoost and LightGBM automated valuation models (AVMs) for each city, measure assessment equity using IAAO ratio statistics stratified by race and income, and simulate a revenue-neutral LVT using actual 2023 Cook County tax rates. Neighborhoods are clustered by bill-change distribution using Gaussian Mixture Models. In Cook County, 61.3% of residential parcels benefit under LVT, but majority-Black neighborhoods are the only racial group below 50% (46.6%). We trace this to high land-to-improvement ratios in historically redlined areas and show that a $10,000 homestead exemption on land assessed value raises all racial groups above 80% benefit while remaining revenue-neutral. Results across cities do not uniformly support LVT as a remedy for racial tax disparities without complementary policy. This analysis covers three cities; extension to additional jurisdictions is left to future work.**

***Index Terms*—property tax regressivity, assessment equity, land-value tax, automated valuation models, XGBoost, LightGBM, racial and income inequality, Cook County, New York City, Philadelphia**

I. Introduction

Property tax liability is determined not by income but by government estimates of property value. Evidence that these estimates are systematically biased is substantial: lower-value properties are routinely assessed at higher fractions of market value than higher-value ones [1]–[3], and majority-Black neighborhoods in cities including Chicago, Detroit, and New Orleans carry effective burdens well above those of comparable white neighborhoods [4], [5]. The nominal rate may be uniform; the effective rate is not.

The Cook County Assessor's Office (CCAO) has addressed this with an unusual degree of transparency, publishing open-source valuation models [6] and treating assessment equity as an explicit organizational objective, providing a replicable baseline for this study.

We examine three questions across Cook County (IL), New York City (NY), and Philadelphia (PA): Do official assessed values systematically over-assess properties in lower-income and majority-Black neighborhoods relative to model-predicted market values? Would a revenue-neutral LVT reduce those effective-rate disparities? And do the answers hold across three cities with different assessment regimes and land markets?

The pipeline has four stages. We train AVMs (OLS, Random Forest, XGBoost, LightGBM) for each city and use the assessment ratio (official assessed value divided by predicted market value) as the primary equity measure. We simulate a revenue-neutral LVT scaled to preserve total collections. We cluster census tracts by their implied tax-change distributions using GMMs. Finally, we compare distributional patterns across cities.

Prior work has examined assessment regressivity [3], [5] and land value taxation [7]–[9] largely separately. This study links both: because LVT shifts burden toward land value relative to improvements, its distributional consequences depend critically on the historical structure of neighborhood property values, which has been shaped by decades of racial discrimination. This work was motivated by a research initiative led by Frederiksen, who hypothesized that revenue-neutral LVT in Cook County would lower taxes for the majority of homeowners. The economic rationale follows Ganong and Shoag [13], who showed that rising housing costs in high-income places have slowed income convergence; LVT theory predicts that taxing land over improvements could counteract speculative holding and incentivize construction [7], [8], though this paper's static analysis cannot measure such behavioral effects.

II. Background

George [7] argued that taxing land rent eliminates deadweight loss from taxing productive activity, since fixed land supply means the tax cannot reduce quantity. Dye and England [9] survey LVT theory and practice across 30+ countries, documenting both efficiency arguments and implementation constraints. Mieszkowski and Zodrow [10] provide the incidence-theory foundation for analyzing how property tax burdens distribute across uses and locations.

Empirical evidence on LVT effects is limited. Plassmann and Tideman [8] used a Markov Chain Monte Carlo analysis of Pennsylvania split-rate cities and found a positive construction effect when land is taxed more heavily than improvements. Pittsburgh's system (1913–2001, land taxed at 5.77 times the

rate on structures [16]) is the canonical example: Oates and Schwab [11] found its 1979–80 restructuring contributed to a commercial building boom well in excess of comparable Rust Belt cities, with the land tax enabling the city to avoid tax increases that would have suppressed development. The system was abandoned in 2001 after deferred reassessments produced an 81% jump in assessed land values from political failure [16].

The distributional stakes of any LVT shift are set by the legacy of housing discrimination. Perry et al. [14] find that majority-Black neighborhood homes are valued 23% below comparable majority-White homes even after controlling for structural and neighborhood characteristics. Pappas [12] documents that Cook County properties in formerly redlined areas are 2.75 times more likely to be distressed today, and 72% of the $\sim$27,000 properties in the county's scavenger sale are in predominantly Black wards, the same improvement-depleted parcels that generate high land-to-structure ratios and determine LVT outcomes.

The CCAO's open-source pipeline [6] uses LightGBM trained on arms-length sales with parcel, neighborhood, and time-of-sale features evaluated against IAAO ratio standards (COD, PRD, PRB) [1]. We adopt its feature conventions across all three cities. Avenancio-León and Howard [5] document the nationwide "assessment gap" (assessments less sensitive to neighborhood characteristics than market prices) and find that the resulting disparity is driven by regressivity in practice rather than direct racial targeting, a mechanism this paper tests and largely confirms in Cook County.

III. Data

Each city contributes assessment records, arms-length sales, and Census tract demographics from the 5-year ACS. Table I summarizes the resulting datasets.

Cook County assessment records come from the CCAO data portal [6] and include parcel values at three appeal stages (mailed, certified, Board of Review), resolved to the most recent non-null value. Building characteristics (square footage, age, bedrooms, bathrooms, garage size, construction quality, exterior wall) come from the assessor's parcel universe. Sales cover 2021–2023, filtered to single-parcel arms-length transactions between \$10,000 and \$10,000,000. Illinois residential properties are assessed at 10% of market value, so assessed values are divided by 0.10 to recover implied market values. Ten spatial proximity features supplement the parcel data: transit distances (CTA, Metra), Lake Michigan distance, water proximity, bus stop and school counts, foreclosure density (5-year), FEMA flood factor, and O'Hare noise contour. After filtering to residential non-exempt parcels with positive assessed values, the Cook County dataset contains 1,635,253 parcels, of which 127,549 have qualifying sales.

New York City parcel data come from the PLUTO dataset (Department of City Planning), including land and total assessed value, building class, gross square footage, lot area, year built, residential units, and floors. Sales come from the Department of Finance rolling files, filtered to \$10,000–\$50,000,000. AVM training and equity analysis are restricted to Tax Class 1 (1–3 family homes; $n = 565{,}301$ parcels, $n = 114{,}754$ qualifying sales), consistent with prior NYC assessment literature. Tax Class 2 properties (apartments, co-ops, condominiums; $n = 139{,}872$) are assessed under the income capitalization method required by RPTL §581, making hedonic AVM ratios inappropriate for that class [17].

Philadelphia parcel data come from the Office of Property Assessment (OPA), including livable area, lot area, year built, bedrooms, bathrooms, rooms, zoning, and building code. OPA assesses at 100% of market value, so no level adjustment is needed. The LVT simulation uses `taxable_total` rather than full market value, consistent with the other cities. Sales are filtered to \$10,000–\$10,000,000 and restricted to 2019 onward for current-market relevance. OPA sets land value using block-level schedules, producing near-zero cross-sectional variation for non-abated parcels. The LVT simulation is therefore driven by the bimodal distribution from Act 76 abatements: abated parcels have `taxable_building` $= 0$ and thus `taxable_land`/`taxable_total` $= 1.0$, while non-abated parcels sit near 0.20–0.25. Of the 504,034 residential parcels, 42,470 (8.4%) were abated with missing building values and imputed at the citywide median land share of 0.20 from non-abated parcels – a modeling assumption that introduces measurement error into both the AVM training set and the LVT simulation for that subset.

Demographics for all three cities come from the 5-year ACS (2019–2023). We retain median household income, racial composition (Black, Hispanic, white shares), and housing tenure, joining parcels to tracts via GEOID. Tracts are classified majority-Black, majority-Hispanic, or majority-White when any group exceeds 50%; income quartiles are computed within each city.

TABLE I
Dataset Summary by City

	Cook County	New York City	Philadelphia
Total parcels	1,893,758	858,644	583,558
Residential	1,635,253	705,173	504,034
AVM training set	127,549	114,754	135,712
Census tract match	79.6%	89.0%	64.0%
Median sale price	\$325K	\$785K	\$228K

IV. Methodology

A. Automated Valuation Models

We train OLS, Random Forest, XGBoost, and LightGBM on the log-transformed sale price (predictions exponentiated back to dollars for evaluation). Numeric features are standardized; categorical features are one-hot encoded via a shared `ColumnTransformer` fit on training data only. Observations outside the 1st/99th percentiles of log-price are dropped. We use an 80/20 split and evaluate on R^2, RMSE, MAE, and MdAPE.

Feature sets differ by city. Cook County: square footage, age, bedrooms, bathrooms, rooms, fireplace count, garage size,

attic finish, township, neighborhood, building type, exterior wall, basement, air conditioning, and construction quality, plus 10 spatial features. NYC (Class 1 only): gross area, lot area, age, residential units, floors, borough, and building class. Philadelphia: livable area, lot area, age, bedrooms, bathrooms, rooms, zip code, zoning, and building code. Cook County's spatial features are unavailable for the other two cities, so cross-city R^2 comparisons are partially confounded by feature availability and should not be taken as reflecting differences in underlying predictability. The OLS AVM is excluded from NYC and Philadelphia results due to numerical instability from outliers in those cities' implied market value distributions.

B. Assessment Equity Metrics

The assessment ratio for parcel i is the official assessed value (scaled to implied market value by the statutory assessment level) divided by the model-predicted market value. Ratios above 1.0 indicate over-assessment; below 1.0, under-assessment. We compute IAAO ratio statistics on the training-set sales: COD (dispersion), PRD (vertical equity), and PRB (regression-based regressivity) [1]. Acceptable ranges are COD 5–15%, PRD 0.98–1.03, PRB ± 0.05; a statistically significant negative PRB ($p < 0.05$) is required to assert regressivity.

For neighborhood-level disparities, we regress the tract-median assessment ratio on percent Black, percent Hispanic, log median household income, log median predicted value, median land-to-total ratio, and percent owner-occupied. We report ratios by racial group and income quartile, and intersectionally. The NYC equity analysis is restricted to Tax Class 1; including Class 2/3 parcels (ratio $= 1.0$ by construction) would mechanically suppress any disparity signal.

C. Land Value Tax Simulation

The LVT replaces the current tax base (land plus improvements) with land value only, scaling the rate so that total collections across all taxable parcels remain constant – what we mean by "revenue neutral" throughout. The multiplier is:

$$m = \frac{\sum_i AV_i^{\text{total}} \times r_i}{\sum_i AV_i^{\text{land}} \times r_i} \tag{1}$$

where sums run over all non-exempt parcels with positive assessed values and r_i is the per-parcel composite rate. Estimated tax bills are:

$$T_i^{\text{current}} = AV_i^{\text{total}} \times e \times r_i \tag{2}$$

$$T_i^{\text{LVT}} = AV_i^{\text{land}} \times e \times r_i \times m \tag{3}$$

where e is the state equalization factor. For Cook County we use parcel-specific rates from the Cook County Clerk's 2023 data [15] (4,464 tax codes, range 1.7–32.1%, median 7.52%) and the official equalization factor $e = 3.0163$. For Philadelphia, $e = 1.0$ and rate $= 1.3998\%$. For NYC, $e = 1.0$ and rate $= 10\%$ (placeholder). The NYC placeholder understates actual FY2023 Class 1 rates by roughly 50% (true rate $\approx 20.31\%$ of assessed value), so all NYC dollar-denominated LVT estimates are understated by that margin; benefit percentages and Gini coefficients, which depend on relative magnitudes, are unaffected. A parcel is coded as an LVT beneficiary if $T_i^{\text{LVT}} < T_i^{\text{current}}$.

D. Neighborhood Clustering

Parcel outcomes are aggregated to census tracts and clustered with GMMs on a feature vector comprising: median assessment ratio, median land-to-total ratio, median percent tax change, share benefiting, median household income, percent Black, percent Hispanic, percent owner-occupied, and median predicted market value. Features are standardized; PCA (5 components, $\geq 90\%$ variance retained) reduces multicollinearity. Component count is selected by BIC.

E. LVT Benefit Classification

We train logistic regression, a linear SVM (wrapped with `CalibratedClassifierCV`), and a two-hidden-layer MLP (128–64 units, ReLU, Adam, early stopping) to predict whether a parcel pays less under LVT. Features are total market value, assessment ratio, building size, age, bedrooms, bathrooms, and tract-level income and racial composition. Land and building market value components are excluded from the Cook County and NYC feature sets: including the land-to-total ratio would directly encode the outcome and make the task trivial. Models are evaluated on accuracy, precision, recall, F1, and AUC.

F. Transition Analysis

For Cook County, we run two supplementary simulations. A phased-transition blends current and LVT tax bills at 20/80, 40/60, 60/40, and 80/20 over five years. A homestead exemption simulation shields the first $X \in \{0, 10{,}000, 20{,}000, 30{,}000, 50{,}000\}$ dollars of land assessed value, recomputing the revenue-neutral rate at each level.

V. Results

A. AVM Performance

LightGBM is the best-performing model in all three cities (Table II). Cook County ($R^2 = 0.817$) approaches the CCAO's production model ($R^2 \approx 0.85$–0.90); the gap reflects their use of 80+ features and post-model adjustments. NYC ($R^2 = 0.572$, MdAPE 15.6%) is applied to Class 1 only; the lower R^2 reflects both market heterogeneity and the absence of spatial features. Philadelphia ($R^2 = 0.735$, MdAPE 20.4%) performs well given the compressed price range.

TABLE II
AVM Performance – LightGBM (best model, all cities)†

City	R^2	RMSE	MAE	MdAPE
Cook County	0.817	$125,659	$77,774	15.2%
New York City	0.572	$340,909	$206,625	15.6%
Philadelphia	0.735	$94,495	$63,454	20.4%

† Cook County includes 10 spatial features unavailable for NYC/Philadelphia; cross-city R^2 is not directly comparable.

B. IAAO Ratio Statistics

All three cities fail IAAO standards on COD and PRD (Table III). Cook County's COD of 28.1% reflects high dispersion the CCAO has acknowledged [6]. NYC's PRB of -1.048 and Philadelphia's -0.492 indicate severe regressivity in both: lower-value properties carry systematically higher assessment ratios. Cook County's PRB (0.015) is within the acceptable range, consistent with the CCAO's equity-focused model. These statistics are computed on arms-length sales, so high COD values partly reflect prediction uncertainty; systematic PRD and PRB failures indicate true bias that persists across groups.

TABLE III
IAAO RATIO STUDY STATISTICS (TRAINING-SET SALES)

City	N	Median Ratio	COD	PRD	PRB
Cook County	196,029	0.884	28.1%*	1.093*	0.015
New York City	114,811	1.150	39.9%*	1.249*	−1.048*
Philadelphia	135,760	1.091	42.8%*	1.255*	−0.492*

* Outside IAAO range. Standards: COD 5–15%, PRD 0.98–1.03, PRB −0.05 to 0.05 [1].

C. Assessment Equity

TABLE IV
MEDIAN ASSESSMENT RATIO BY RACIAL GROUP

Group	Cook County	NYC (Class 1)	Philadelphia
Majority Black	0.630	1.048	1.213
Majority Hispanic	0.776	1.165	1.479
Mixed	0.764	1.230	1.205
Majority White	0.816	1.329	1.159

In Cook County, majority-Black tracts have a median ratio of 0.630 versus 0.816 for majority-White, an 18.6 pp gap (Table IV). The `pct_black` coefficient in the tract OLS is -0.0002 ($t = -0.874$, $p = 0.382$), not significant. The model controls for features that are themselves products of segregation: transit access, foreclosure density, flood risk, and noise. That race adds no independent explanatory power after these controls suggests Cook County's assessment disparities operate through place-based disinvestment rather than direct racial targeting, consistent with Avenancio-León and Howard [5]. Majority-Hispanic tracts are over-assessed after spatial controls ($\beta = 0.0013$, $p < 0.001$); higher-income tracts have ratios closer to 1.0 ($\beta = 0.110$, $p < 0.001$). $R^2 = 0.339$ across 1,270 tracts.

In NYC, the Class 1 analysis covers 1,576 tracts. Majority-Black tracts have a median ratio of 1.048 versus 1.329 for majority-White (28.1 pp). The OLS confirms significant racial disparities: `pct_black` $= -0.0044$ ($t = -15.3$, $p < 0.001$) and `pct_hispanic` $= -0.0013$ ($t = -2.8$, $p = 0.005$). The positive `log_predicted_value` coefficient ($\beta = 0.346$, $p < 0.001$) is the inverse of the usual regressivity pattern and reflects NYC's Class 1 statutory cap mechanism: lower-value homes that appreciated rapidly are constrained below the 6% target, while higher-value properties in slower-appreciating areas track closer to it. The racial disparity is therefore driven by differential cap exposure rather than assessor undervaluation of minority-neighborhood properties, a structural mechanism documented by the NYC Advisory Commission [17]. $R^2 = 0.399$ across 1,576 tracts.

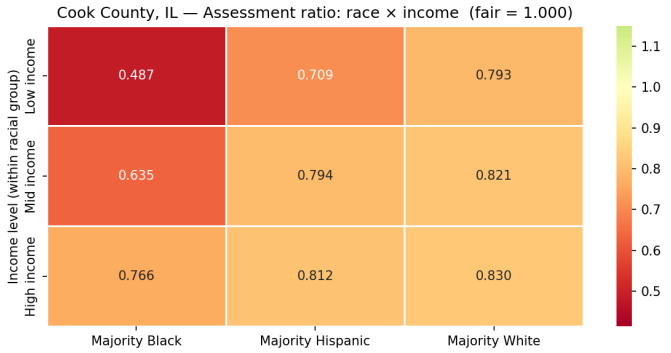

Fig. 1. Cook County assessment ratios by race and income tercile. Low-income Black tracts (0.48) face a larger gap than low-income White tracts (0.79), indicating the racial disparity persists within income brackets.

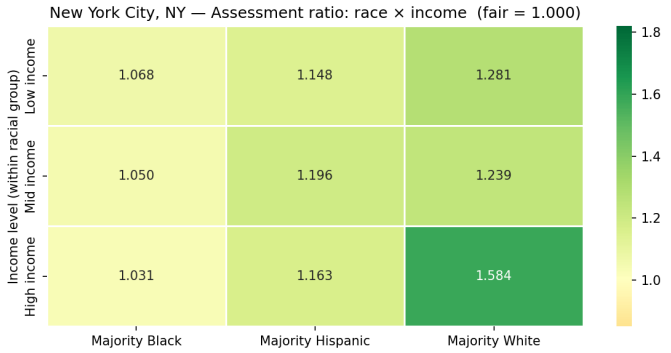

Fig. 2. NYC Class 1 assessment ratios by race and income tercile. Low-income Black tracts have the lowest ratios, consistent with cap constraints on appreciating minority-neighborhood properties.

Philadelphia ratios are uniformly above 1.0 (1.159–1.479), indicating city-wide over-assessment relative to model predictions. The Black-White gap is only 5.4 pp, and neither `pct_black` ($p = 0.287$), `pct_hispanic` ($p = 0.150$), nor `log_income` ($p = 0.487$) is significant. $R^2 = 0.163$ across only 245 tracts. The minimum detectable effect at 80% power (0.0034 pp per unit of `pct_black`) would be sufficient to detect the Hispanic-White raw gap of 0.320 if it were systematic; the null is not purely a power problem. That said, the 64% census tract match rate limits confidence.

D. LVT Simulation

In Cook County, 61.3% of residential parcels would pay less under revenue-neutral LVT (Table V). Majority-Black neighborhoods benefit at 46.6(0.396 to 0.433), meaning effective-rate dispersion widens despite the majority benefiting. The driver is GMM cluster 4 (75% Black, median land ratio 0.45, 96% median tax increase, 11% LVT benefit rate): disinvested parcels with high land-to-improvement ratios in historically

TABLE V
LVT SIMULATION RESULTS‡

	Cook County	New York City	Philadelphia
LVT multiplier	4.40×	5.67×	3.55×
Revenue deviation	<0.01%	<0.01%	<0.01%
Overall % benefit	61.3%	30.3%	58.2%
% benefit (Maj. Black)	46.6%	18.8%	52.3%
% benefit (Maj. White)	64.2%	35.2%	64.2%
Gini current	0.396	0.444	0.279
Gini LVT	0.433	0.458	0.186

‡ Revenue deviation is floating-point rounding only.
NYC dollar estimates understated ≈50%; see Sec. IV-C.
Gini not comparable across cities (different AVM scales).

TABLE VI
CLASSIFIER PERFORMANCE – LVT BENEFIT PREDICTION

City	Model	Acc.	Prec.	Rec.	F1	AUC
Cook County	Log. Reg.	0.741	0.762	0.801	0.795	0.857
	SVM (Linear)	0.736	0.758	0.792	0.789	0.851
	Neural Net	0.851	0.866	0.922	0.894	0.959
New York City	Log. Reg.	0.712	0.348	0.421	0.381	0.765
	SVM (Linear)	0.709	0.345	0.428	0.382	0.750
	Neural Net	0.741	0.621	0.738	0.678	0.875
Phil-adel-phia	Log. Reg.	0.703	0.731	0.774	0.752	0.816
	SVM (Linear)	0.700	0.729	0.770	0.749	0.813
	Neural Net	0.730	0.758	0.817	0.787	0.870

redlined areas. Per-parcel rate data confirm that majority-Black south suburban communities (Ford Heights, Phoenix, Riverdale, Harvey) hold the highest rates in the county and face the sharpest LVT transition costs.

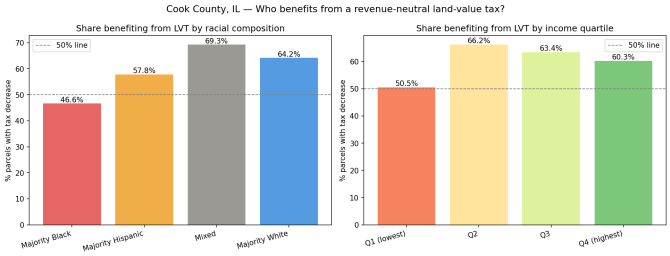

Fig. 3. Cook County share of parcels paying less under LVT, by racial composition (left) and income quartile (right). Majority-Black neighborhoods are the only group below 50%; lower-income quartiles benefit less, reflecting high land-to-improvement ratios in disinvested areas.

In NYC, only 30.3% benefit overall, with majority-Black at 18.8% versus 35.2% for majority-White. The Gini rises modestly (0.444 to 0.458). The 5.67× multiplier reflects the high land fraction of assessed values in a land-scarce market. The LVT simulation runs on all 705,173 parcels using `assesstot`/`assessland` directly and is not affected by the Class 1 AVM restriction.

Philadelphia shows 58.2% benefit overall (majority-Black 52.3%, majority-White 64.2%); the Gini falls from 0.279 to 0.186. The result is mechanically driven by abatement status: abated parcels have `taxable_building` = 0, so `taxable_land`/ `taxable_total` = 1.0, guaranteeing a tax cut. This is a valid policy counterfactual (what if Philadelphia taxed only the recorded taxable land component), but it is not equivalent to a LVT based on appraised land values.

E. LVT Benefit Classification

Land and building market value components are excluded from the classifier feature sets; including the land-to-total ratio would directly encode the outcome. Models must instead infer LVT benefit status from building characteristics and neighborhood demographics.

The neural network leads in all three cities (Table VI). AUC gaps over logistic regression are 10.2 pp in Cook County (0.959 vs. 0.857), 11.0 pp in NYC (0.875 vs. 0.765), and 5.4 pp in Philadelphia (0.870 vs. 0.816). The consistency of this gap across independent datasets with different feature sets indicates genuine nonlinearity: linear classifiers miss interactions between building age, lot size, and neighborhood composition that the neural network captures. Philadelphia, with a near-balanced 52.3% benefit rate, provides the most informative test; an AUC of 0.870 demonstrates that building and demographic features predict LVT outcomes accurately even without land value data.

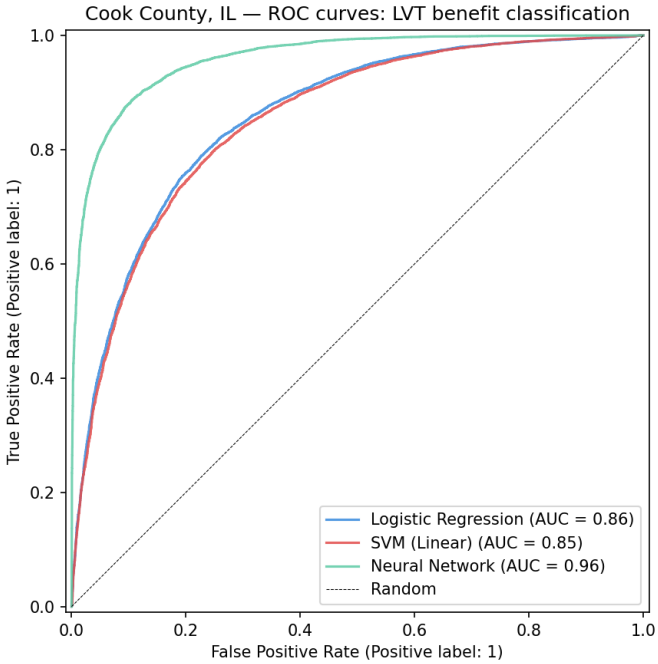

Fig. 4. Cook County ROC curves (land-value features excluded). Neural network AUC 0.959 versus logistic regression 0.857.

F. Neighborhood Clustering

GMM identified 8 clusters in Cook County (silhouette 0.144), 8 in NYC (0.195), and 5 in Philadelphia (0.277). Low silhouette scores in Cook County and NYC indicate tract outcomes form a continuum rather than discrete groups. At $k = 2$, Cook County splits into a majority-White/ Hispanic cluster (940 tracts, 3.8% Black, $88K median income, 62.5% LVT benefit) and a majority-Black cluster (330 tracts, 85.8%

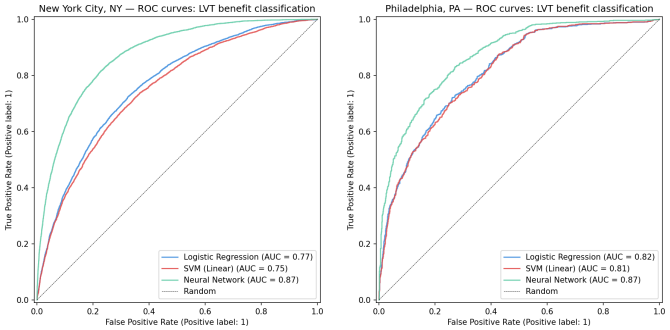

Fig. 5. ROC curves for NYC (left) and Philadelphia (right). AUC gaps of 11.0 and 5.4 pp over logistic regression.

Black, $47K income, 35.0% LVT benefit), despite no racial variables in the input. This unsupervised recovery of racial geography from property data alone is consistent with the hypothesis that segregation has embedded itself in land-to-improvement ratios, though the clustering cannot establish causation.

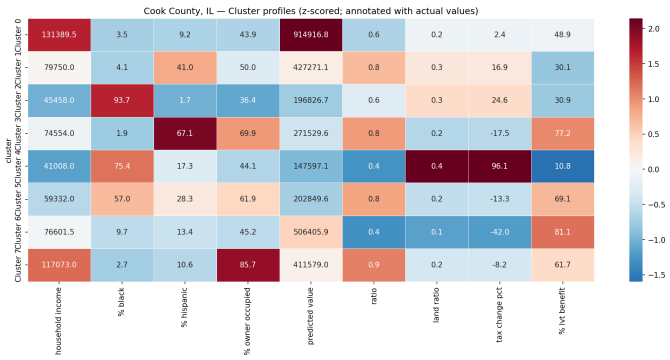

Fig. 6. Cook County GMM cluster profiles (z-scored, actual medians annotated). Clusters 4 and 6 (majority-Black, high land ratio) show median tax increases of +96% and +24% under LVT. Cluster 3 (majority-Hispanic, high homeownership) is the primary beneficiary.

G. Transition Analysis

The phased-transition simulation shows majority-Black neighborhoods in Cook County stabilizing near 46% LVT benefit at every phase; they never cross 50%. Phasing delays but does not resolve the structural outcome.

TABLE VII
HOMESTEAD EXEMPTION IMPACT ON LVT BENEFIT RATE (COOK COUNTY)[§]

Exemption	Maj. Black	Maj. Hispanic	Maj. White
$0 (pure LVT)	46.6%	57.8%	64.2%
$10,000	98.2%	95.7%	83.8%
$20,000	99.7%	99.8%	94.1%
$30,000	99.9%	100.0%	96.8%
$50,000	100.0%	100.0%	99.3%

[§] $0 row matches Table V. Rate recomputed at each level to maintain revenue neutrality.

A $10,000 homestead exemption on land assessed value raises the majority-Black benefit rate from 46.6% to 98.2% (Table VII). Land assessed values in formerly redlined neighborhoods are typically $6–8K at the 10% residential assessment level, so the exemption shields most Black homeowners entirely while high-value land and speculative vacant lots continue to bear the full LVT burden.

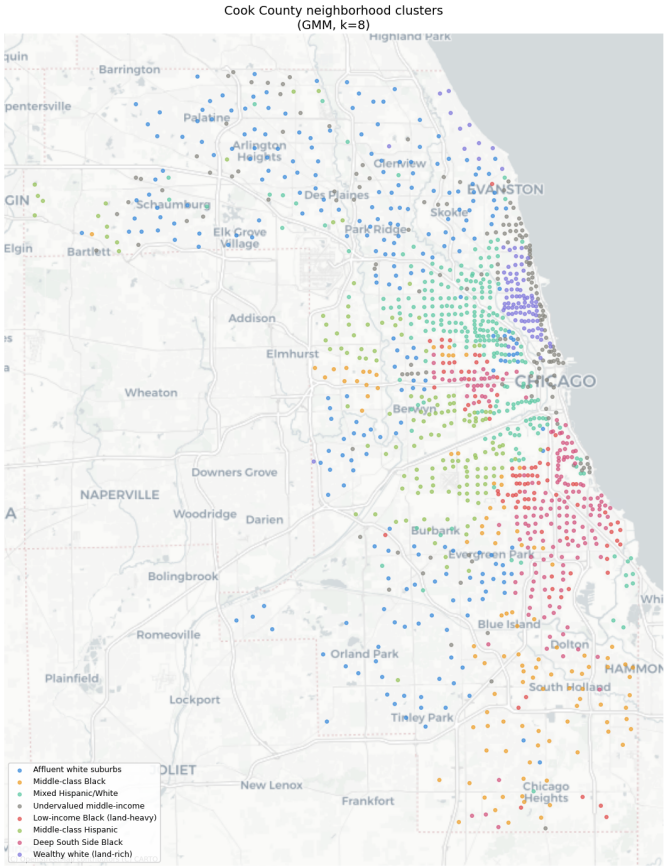

Fig. 7. Cook County GMM clusters by census tract centroid. North Shore high-value tracts (cluster 0), South and West Side majority-Black tracts (clusters 4–5), and suburban rings with varied outcomes are spatially distinct.

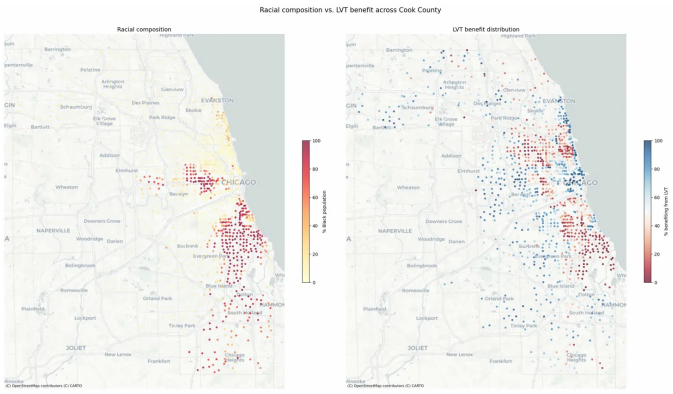

Fig. 8. Cook County: Black population share (left) and LVT benefit distribution (right).

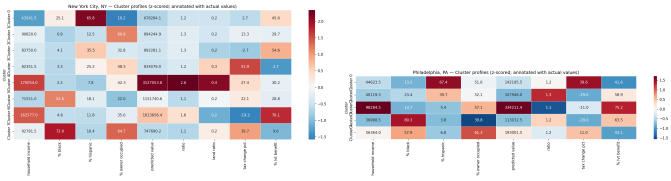

Fig. 9. GMM profiles for NYC (8 clusters, left) and Philadelphia (5 clusters, right). NYC cluster 4 (44 tracts, median value \$3.1M, median ratio 2.57) captures ultra-high-value Class 1 properties. Philadelphia cluster 0 ($n = 4$) is a likely artifact.

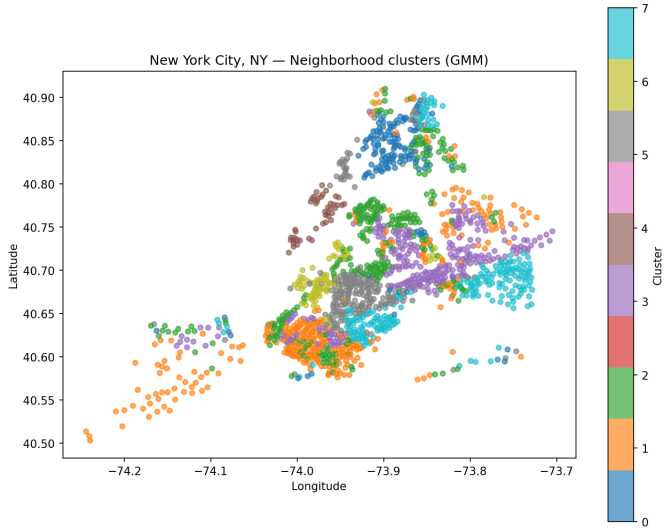

Fig. 10. NYC GMM clusters by census tract centroid. Majority-Black Brooklyn and Bronx tracts (clusters 5 and 7) are spatially distinct from high-value Manhattan and Staten Island clusters.

VI. Discussion

The Black-White benefit gap under LVT is 17.6 pp in Cook County (46.6% vs. 64.2%), 16.4 pp in NYC (18.8% vs. 35.2%), and 11.9 pp in Philadelphia (52.3% vs. 64.2%). In no city does the Black benefit rate exceed the overall rate. This consistency across cities with different assessment regimes, housing markets, and demographic compositions points to a structural rather than idiosyncratic cause: majority-Black neighborhoods carry higher land-to-improvement ratios than majority-White neighborhoods as a direct consequence of redlining, mortgage discrimination, and decades of disinvestment.

The IAAO results support regressivity claims in two of the three cities. Cook County's PRB (0.015, within range) and the insignificant OLS racial coefficient, once spatial disadvantage controls absorb the racial signal, suggest disparities operate through place rather than assessor targeting. NYC's PRB (-1.048, severe) and strongly significant racial OLS coefficients confirm systematic disparities, though the mechanism is the Class 1 statutory cap rather than direct under-assessment: lower-value appreciating homes in minority neighborhoods are constrained below the 6% target, documented by the NYC Advisory Commission [17] and confirmed here. Philadelphia shows no statistically significant racial disparity in its equity OLS, though the 245-tract sample and 64% match rate limit any strong negative conclusion.

On the LVT question: Cook County's Gini *increases* under LVT (0.396 to 0.433), meaning effective-rate dispersion widens even as a majority of homeowners benefit. The clustering analysis traces this to GMM clusters 4 and 6, where 75% of residents are Black, land ratios are high, and median tax increases under LVT are $+96\%$ and $+24\%$. Phasing the transition over five years does not resolve this: majority-Black tracts never cross the 50% benefit threshold. The homestead exemption does. At \$10,000, it raises the Black benefit rate from 46.6% to 98.2% while maintaining revenue neutrality, because the same land assessed values that generate LVT losses (\$6–8K in redlined neighborhoods) fall below the exemption threshold, shifting the residual burden entirely onto high-value and speculative holdings.

Philadelphia's Gini improvement (0.279 to 0.186) is a valid policy counterfactual but does not reflect true land value heterogeneity: OPA's block-level schedules assign $\approx 20\%$ land share to essentially all non-abated parcels, so the simulation is driven by abatement-status bimodality rather than appraised land values. Results should be interpreted accordingly. The classification finding, a consistent 5–11 pp AUC advantage for the neural network over linear models across all three cities, indicates that the relationship between observable parcel and neighborhood characteristics and LVT benefit status is genuinely nonlinear. GMM with EM, preferred over k-means, accommodates full covariance structure and soft assignments, appropriate given the continuous gradients observed in Cook County and NYC (silhouettes 0.144 and 0.195).

VII. Limitations

Parcel-level assessment ratios carry 15.2–20.4% median prediction error (city-dependent). The equity analysis aggregates to tract medians, where errors are approximately mean-zero, but IAAO failures on COD and PRD (Table III) introduce uncertainty into the tract-level OLS. All three cities fail those standards, confirming that the observed disparities exceed what prediction error alone can produce.

The Cook County simulation uses a single county-wide LVT multiplier; per- district multipliers would produce different patterns in municipalities with extreme rates (1.7–32.1%), particularly the majority-Black south suburban townships that already bear the highest rates and the sharpest LVT costs.

NYC AVM training is restricted to Class 1 ($n = 114{,}754$ sales) per RPTL §581's income-capitalization requirement for Class 2. The 139,872 Class 2/3 parcels are assigned ratio $= 1.0$ by convention, suppressing any disparity signal for that population.

Philadelphia's 64% census tract match means 36% of parcels lack demographic covariates; the equity OLS on 245 tracts is the smallest sample in the study, and the null racial result is not strong evidence of equity. The LVT result is driven by abatement bimodality, not land value heterogeneity.

Renter impacts are unmodeled. Property tax increases are passed through to rents, and renters (disproportionately people of color) cannot directly benefit from LVT tax decreases; this omission likely understates racial equity costs across all three cities.

The analysis is static and cross-sectional. It does not capture behavioral responses (construction, relocation, rent adjustment, speculative holding) that would follow actual LVT enactment. Extension to additional cities, longitudinal analysis across reassessment cycles, and behavioral incidence modeling are the natural next steps.

VIII. Conclusion

Assessment disparities along racial and income lines are present in Cook County (Black-White ratio gap 18.6 pp; PRB within range, operating through spatial disadvantage) and New York City (28.1 pp gap; PRB $= -1.048$, driven by cap-mechanism differential exposure). Revenue-neutral LVT would lower taxes for majorities of residential parcels in Cook County (61.3%) and Philadelphia (58.2%), but only 30.3% in NYC. In no city does LVT close the Black-White benefit gap, and in Cook County majority-Black neighborhoods are the only racial group below 50% benefit.

A $10,000 homestead exemption on Cook County land assessed value raises the majority-Black benefit rate from 46.6% to 98.2% while remaining revenue-neutral. Any property tax policy that does not account for the land-to-improvement structures that discrimination created will reproduce the inequities it nominally aims to correct.

Acknowledgments

This work was supported by a research initiative led by James Frederiksen (Duke University). We thank the Cook County Assessor's Office, Clerk's Office, and Treasurer's Office for their commitment to open data. Course: MATH 7339, Machine Learning and Statistical Learning Theory II, Northeastern University (Instructor: He Wang).

References

[1] International Association of Assessing Officers (IAAO), "Standard on Ratio Studies," IAAO, Kansas City, MO, 2013.

[2] T. R. Hodge, D. P. McMillen, G. C. Sands, and M. Skidmore, "Assessment Inequity in a Declining Housing Market: The Case of Detroit," *Real Estate Economics*, vol. 45, no. 2, pp. 237–258, 2017.

[3] C. Berry, "Reassessing the Property Tax," University of Chicago Harris School of Public Policy, Working Paper, 2021.

[4] ProPublica Illinois, "The Tax Divide: An Investigation into Inequality in Chicago's Property Tax System," 2017. [Online]. Available: https://www.propublica.org/series/the-tax-divide

[5] C. Avenancio-León and T. Howard, "The Assessment Gap: Racial Inequalities in Property Taxation," *The Quarterly Journal of Economics*, vol. 137, no. 3, pp. 1383–1434, 2022.

[6] Cook County Assessor's Office Data Science Team, "CCAO Modeling Repository," GitHub, 2021. [Online]. Available: https://github.com/ccao-data

[7] H. George, *Progress and Poverty*. New York: D. Appleton and Company, 1879.

[8] F. Plassmann and T. N. Tideman, "A Markov Chain Monte Carlo Analysis of the Effect of Two-Rate Property Taxes on Construction," *Journal of Urban Economics*, vol. 47, no. 2, pp. 216–247, 2000.

[9] R. F. Dye and R. W. England, *Assessing the Theory and Practice of Land Value Taxation*. Cambridge, MA: Lincoln Institute of Land Policy, 2010.

[10] P. Mieszkowski and G. R. Zodrow, "Taxation and the Tiebout Model: The Differential Effects of Head Taxes, Taxes on Land Rents, and Property Taxes," *Journal of Economic Literature*, vol. 27, no. 3, pp. 1098–1146, 1989.

[11] W. E. Oates and R. M. Schwab, "The Impact of Urban Land Taxation: The Pittsburgh Experience," *National Tax Journal*, vol. 50, no. 1, pp. 1–21, 1997.

[12] M. Pappas, "Maps of Inequality: From Redlining to Urban Decay and the Black Exodus," Cook County Treasurer's Office, July 2022.

[13] P. Ganong and D. Shoag, "Why Has Regional Income Convergence in the U.S. Declined?" *Journal of Urban Economics*, vol. 102, pp. 76–90, 2017.

[14] A. Perry, J. Rothwell, and D. Harshbarger, "The Devaluation of Assets in Black Neighborhoods," Brookings Institution, 2018.

[15] Cook County Clerk's Office, "2023 Cook County Tax Rates Released," July 2024.

[16] M. Maciag, "Leaning on the Land," *Governing*, April 2021.

[17] New York City Advisory Commission on Property Tax Reform, "Preliminary Report," January 2020. [Online]. Available: https://www.nyc.gov/assets/propertytaxreform/downloads/pdf/NYC-AdvCommission-Prelim.pdf